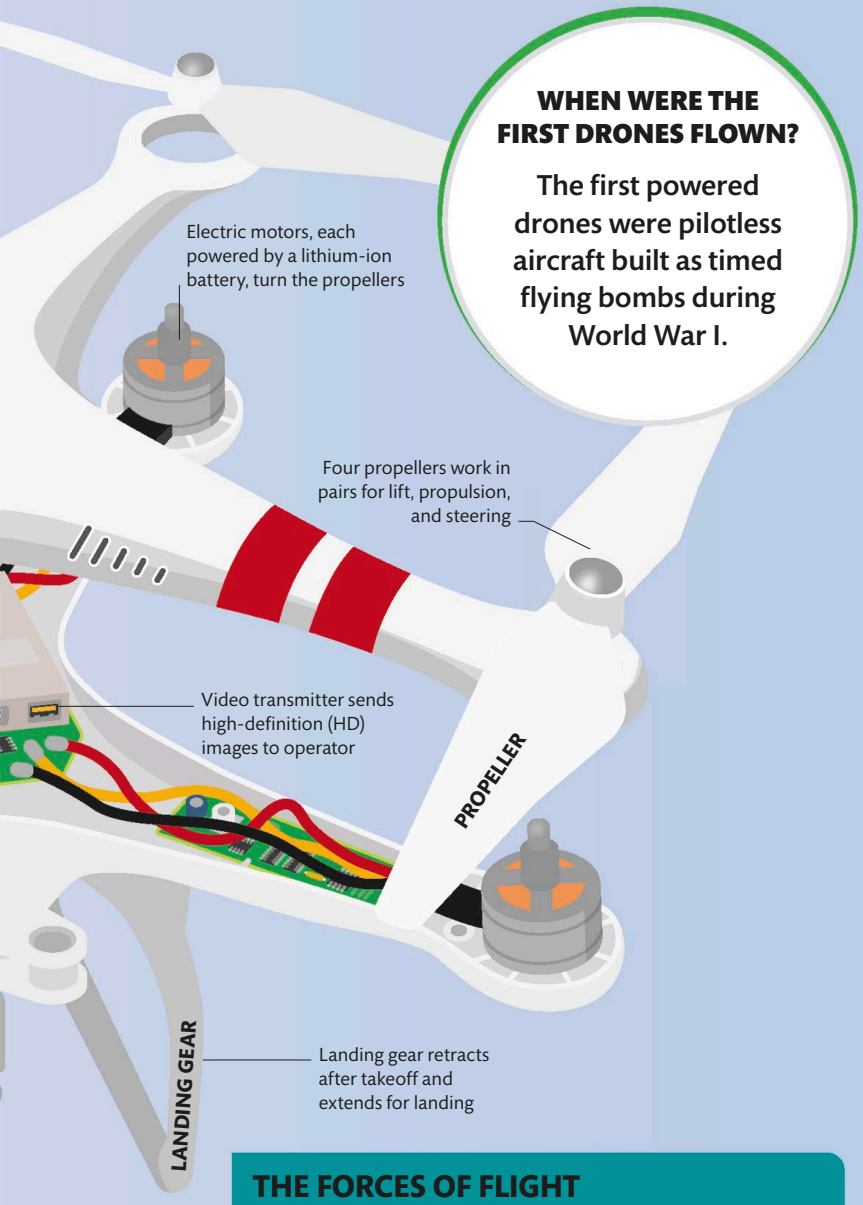
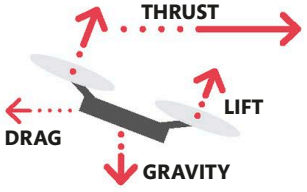


WHEN WERE THE FIRST DRONES FLOWN?

The first powered drones were pilotless aircraft built as timed flying bombs during World War I.

THE FORCES OF FLIGHT

Drones achieve a balance between the four forces of flight (see p.38). Lift and thrust are produced by propellers. They work against gravity and drag respectively to produce vertical and horizontal movement.



Uses of drones

A drone's ability to take off and land almost anywhere, as well as hover at a fixed point over the ground, makes it ideal for a wide range of applications, including surveillance, aerial photography, scientific research, mapmaking, and filming. Broadcasters use drones to capture aerial views of events; farmers use them to assess the health of their crops (see p.220); archaeologists use drones to monitor, map, and protect their sites; and wildlife organizations use them to help protect animals from poachers.



Surveying
Drones can take aerial photographs to map sites faster than traditional methods on the ground.



Military use
Drones flown from very remote distances are used for surveillance, intelligence, and attack without risk to a pilot.



Disaster relief
Medical equipment and medicines can be delivered by drones when land transportation is impractical.



Search and rescue
Some drones are used in search-and-rescue missions. They can deliver equipment to inaccessible places.



Delivery
Courier companies have begun to trial drones for delivering packages weighing up to 4½ lbs (2 kg).



Underwater exploration
Most drones are aircraft, but the term is also used for unmanned underwater craft used for research purposes.